# Assignment Report

## SFWRENG 3S03 (2026W)

## Assignment 1


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GitHub URL

`https://github.com/OmPatel03/3S03-Assignment-1`

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Original template credits: Samuel Zurowski and Charles Barone. Adapted for McMaster SFWRENG by Dr. Istvan David.

# 1 Task 1

## 1.1 (a)

**Prompt + generative-AI response (three types of tests). Prompt used (ChatGPT):** "List three types of tests you would run for an action-adventure game set in post-apocalyptic McMaster University with up to 8 players (co-op + competitive), releasing on PS5, Xbox Series X, Nintendo Switch, and Sega Genesis (16-bit). Keep it practical and explain each test type briefly."

**Response from ChatGPT:**

1. **Platform Compatibility / Certification Testing**
   Verify the game behaves correctly on each platform's hardware, OS, controllers, save system, and platform rules. Special focus for Sega Genesis: memory limits, sprite/tile constraints, audio channels, input latency, and ensuring the game still fails gracefully (e.g., sprite overflow handling) rather than crashing.

2. **Multiplayer / Networking & Sync Testing**
   Test 1–8 players across co-op and competitive modes for state synchronization (positions, damage, AI states), latency tolerance, reconnect behavior, matchmaking/lobbies, host migration (if used), and ensuring no desyncs or soft-locks during chaotic encounters.

3. **Core Gameplay Functional + Regression Testing**
   Validate core loops: movement/collision, combat, enemy AI behaviors (zombies/professors/deans/squirrels), quest/progression triggers, inventory, checkpoints, save/load, and UI. Add regression tests to ensure fixes do not break previously working mechanics.

## 1.2 (b)

**Two poor/unclear things in (a), and why they are unhelpful.**

- **Too broad to be directly actionable.** The test types do not specify concrete pass/fail criteria (for example, what exact result proves sync is correct, or what graceful failure means on Genesis), so it is difficult to convert them into immediate executable test cases.

- **Genesis is mentioned but not operationalized.** The response correctly notes memory and sprite constraints, but it does not define test setup details (scenes, sprite counts, resource budgets, acceptable failure behavior), making the highest-risk platform guidance too abstract.

## 1.3 (c)

**Two good/helpful things in (a), and why they are helpful.**

- **It targets the biggest risk areas for this game.** Cross-platform behavior, multiplayer synchronization, and core gameplay stability are typically where severe and

costly defects appear, so these categories provide a strong high-level strategy.

- **It recognizes Sega Genesis as a special case.** Highlighting 16-bit constraints is useful because it makes clear that test design cannot treat all platforms identically and requires platform-specific validation priorities.

## 1.4 (d)

**Three specific tests, plus why these are preferred.**

1. **Genesis worst-case sprite/tile budget scenario.**
   *Setup:* Execute a stress scene on Sega Genesis with 8 enemies, player combat effects, and UI elements simultaneously visible to push sprite limits.
   *Pass criteria:* No crash or soft-lock occurs, degradation is predictable (e.g., controlled sprite flicker or effect culling), and controls remain responsive.

2. **8-player desync/disconnect robustness.**
   *Setup:* Run an 8-player session, trigger a high-intensity combat sequence, force one player to disconnect mid-combat, then reconnect after 30–60 seconds.
   *Pass criteria:* Ongoing players continue without disruption, and the returning player's position/HP/inventory/objectives match the authoritative state without rubber-banding or broken quest progression.

3. **Save/load state integrity during combat.**
   *Setup:* During an encounter, save at a checkpoint (or manual save), immediately reload, and inspect the full world state.
   *Pass criteria:* Player stats, ammo, enemy states, quest flags, and location restore correctly with no duplication/loss and no incorrect progression reset.

These tests are preferred because each one has a clear setup and measurable pass/fail outcome rather than only broad categories. They also directly target the highest-risk areas in this scenario: Genesis hardware constraints, multiplayer instability, and progression/state integrity.

# 2 Task 2

## 2.1 Program 1: `findLast`

### 2.1.1 (a)

**Fault + precise fix. Fault:** The loop stops at `i > 0`, so index `0` is never checked.
**Fix:** Change the loop condition to include index `0`: `for (int i = x.length - 1; i >= 0; i-)`.

### 2.1.2 (b)

**Test that does not execute the fault.** Use `x = null`, `y = any value`. This throws `NullPointerException` at `x.length`, so the loop condition is not evaluated.

### 2.1.3 (c)

**Test that executes the fault but no error state.** Example: `x = [2,3,5], y = 3, expected = 1`. The faulty condition exists, but the method returns at index `1`, so index `0` is not needed and behavior matches the specification.

### 2.1.4 (d)

**Error state but not failure.** Example: `x = [9,3,5], y = 2, expected = -1`. The error state occurs when `i` becomes `0` and `i > 0` evaluates false, causing loop exit even though the correct version would still execute one more iteration; output remains `-1`, so there is no failure.

### 2.1.5 (e)

**First error state for the given failing test.** Given `x = [2,3,5], y = 2` (expected `0`), the first error state occurs when `i` reaches `0` and `i > 0` evaluates false, so control exits the loop and returns `-1` without checking `x[0]`.

## 2.2 Program 2: `lastZero`

### 2.2.1 (a)

**Fault + precise fix. Fault:** The method returns the first zero because it scans left-to-right and returns immediately.
**Fix:** Scan from the end: `for (int i = x.length - 1; i >= 0; i-) { if (x[i] == 0) return i; } return -1;`.

### 2.2.2 (b)

**Test that does not execute the fault.** Example: `x = [1,2,3], expected = -1`. The faulty `return i;` inside the `if` is never executed because no zero exists.

### 2.2.3 (c)

**Test that executes the fault but no error state.** Example: `x = [1,2,0], expected = 2`. The method returns at index `2`, which is genuinely the last zero, so no incorrect state is reached.

### 2.2.4 (d)

**Error state but not failure.** Example: `x = [1,0,2], expected = 1`. At `i = 1`, control returns immediately without checking later indices, which is a control-state error for a `find last` computation; because no later zero exists, the returned value is still correct.

#### 2.2.5 (e)

**First error state for the given failing test.** Given `x = [0,1,0]` (expected `2`), the first error state is at `i = 0` when `x[0] == 0` causes immediate `return 0`, committing before checking later indices where another zero exists.

### 2.3 Program 3: `countPositive`

#### 2.3.1 (a)

**Fault + precise fix. Fault:** The condition `x[i] >= 0` counts zero as positive.
**Fix:** Use strict positivity: `if (x[i] > 0) { count++; }`.

#### 2.3.2 (b)

**Test that does not execute the fault.** Example: `x = [], expected = 0`. The loop body does not run, so `x[i] >= 0` is never executed.

#### 2.3.3 (c)

**Test that executes the fault but no error state.** Example: `x = [-4,2,5], expected = 2`. No zeros appear, so `>= 0` behaves the same as `> 0` for all elements in this test.

#### 2.3.4 (d)

**Error state but not failure.** Not possible. If the fault is activated (a `0` appears), `count` increments when it should not, which changes the returned value and therefore causes a failure.

#### 2.3.5 (e)

**First error state for the given failing test.** Given `x = [-4,2,0,2]` (expected `2`), the first error state occurs at index `i = 2`: `x[i] >= 0` is true for `0`, so `count` becomes `2` when it should still be `1`.

### 2.4 Program 4: `oddOrPos`

#### 2.4.1 (a)

**Fault + precise fix. Fault:** The check `x[i] % 2 == 1` fails for negative odd numbers in Java (e.g., `-3 % 2 == -1`).
**Fix:** Use `x[i] % 2 != 0 || x[i] > 0`.

#### 2.4.2 (b)

**Test that does not execute the fault.** Example: `x = [], expected = 0`. The loop body does not execute, so the faulty odd-check expression is never evaluated.

#### 2.4.3 (c)

**Test that executes the fault but no error state.** Example: `x = [-2,0,1,4], expected = 2`. No negative odd values appear, so behavior stays consistent with the specification.

#### 2.4.4 (d)

**Error state but not failure.** Not possible. When a negative odd value appears, the method fails to count it, reducing `count`; that incorrect internal state directly propagates to the returned value.

#### 2.4.5 (e)

**First error state for the given failing test.** Given `x = [-3,-2,0,1,4]` (expected `3`), the first error state occurs at `i = 0`: `-3` should be counted as odd, but `-3 % 2 == 1` is false, so `count` remains `0` when it should be `1`.

# 3 Task 3: Testing parts of large systems

## 3.1 (a)

The agent's behaviour can be decoupled using **Dependency Injection** and **Test Doubles** (specifically, stubs/fakes) behind the provided `CatanAgent` interface.

The core engine should solely rely on:

- game rules/state transitions, and
- the contract `CatanAgent` (inputs/outputs)

This allows testing the core module with deterministic, pre-scripted agent choices, so tests focus on legality checks and rule execution rather than RL quality. It also makes later replacement easy: random, heuristic, RL, or human-network agents can all plug in without changing core logic.

## 3.2 (b)

The following classes/interfaces would be implemented to achieve this design:

- `CatanAgent.java`

  ```
  package catan;

  import java.util.List;
  import java.util.Map;

  public interface CatanAgent {
      void init(int playerId);
  ```

```
    Move chooseInitialSettlement(GameState state);

    Move chooseInitialRoad(GameState state);

    Move chooseMove(GameState state);

    Map<ResourceType, Integer> chooseDiscard(GameState state, int discardCount);

    ResourceType chooseResource(GameState state);

    int chooseRobberTarget(GameState state, List<Integer> possibleTargets);

    DevelopmentCard chooseDevelopmentCard(GameState state);
}
```

- AgentPolicy.java

```
package catan;

import java.util.List;
import java.util.Map;

public interface AgentPolicy {
    Move chooseInitialSettlement(GameState state);

    Move chooseInitialRoad(GameState state);

    Move chooseMove(GameState state);

    Map<ResourceType, Integer> chooseDiscard(GameState state, int discardCount);

    ResourceType chooseResource(GameState state);

    int chooseRobberTarget(GameState state, List<Integer> possibleTargets);

    DevelopmentCard chooseDevelopmentCard(GameState state);
}
```

- PolicyBackedCatanAgent.java

```
package catan;

import java.util.List;
import java.util.Map;
import java.util.Objects;
```

```
public final class PolicyBackedCatanAgent implements CatanAgent {
    private final AgentPolicy policy;
    private int playerId;

    public PolicyBackedCatanAgent(AgentPolicy policy) {
        this.policy = Objects.requireNonNull(policy);
        this.playerId = -1;
    }

    @Override
    public void init(int playerId) {
        this.playerId = playerId;
    }

    public int playerId() {
        return playerId;
    }

    @Override
    public Move chooseInitialSettlement(GameState state) {
        return policy.chooseInitialSettlement(state);
    }

    @Override
    public Move chooseInitialRoad(GameState state) {
        return policy.chooseInitialRoad(state);
    }

    @Override
    public Move chooseMove(GameState state) {
        return policy.chooseMove(state);
    }

    @Override
    public Map<ResourceType, Integer> chooseDiscard(GameState state, int discardCou
        return policy.chooseDiscard(state, discardCount);
    }

    @Override
    public ResourceType chooseResource(GameState state) {
        return policy.chooseResource(state);
    }

    @Override
```

```
    public int chooseRobberTarget(GameState state, List<Integer> possibleTargets) {
        return policy.chooseRobberTarget(state, possibleTargets);
    }

    @Override
    public DevelopmentCard chooseDevelopmentCard(GameState state) {
        return policy.chooseDevelopmentCard(state);
    }
}
```

- ScriptedPolicy.java

```
package catan;

import java.util.ArrayDeque;
import java.util.Collections;
import java.util.EnumMap;
import java.util.List;
import java.util.Map;
import java.util.Queue;

public final class ScriptedPolicy implements AgentPolicy {
    private final Queue<Move> scriptedMoves;

    public ScriptedPolicy(List<Move> scriptedMoves) {
        this.scriptedMoves = new ArrayDeque<>(scriptedMoves);
    }

    @Override
    public Move chooseInitialSettlement(GameState state) {
        return nextOrPass();
    }

    @Override
    public Move chooseInitialRoad(GameState state) {
        return nextOrPass();
    }

    @Override
    public Move chooseMove(GameState state) {
        return nextOrPass();
    }

    @Override
    public Map<ResourceType, Integer> chooseDiscard(GameState state, int discardCou
```

```
        return Collections.unmodifiableMap(new EnumMap<>(ResourceType.class));
    }

    @Override
    public ResourceType chooseResource(GameState state) {
        return ResourceType.BRICK;
    }

    @Override
    public int chooseRobberTarget(GameState state, List<Integer> possibleTargets) {
        return possibleTargets.isEmpty() ? -1 : possibleTargets.get(0);
    }

    @Override
    public DevelopmentCard chooseDevelopmentCard(GameState state) {
        return DevelopmentCard.KNIGHT;
    }

    private Move nextOrPass() {
        return scriptedMoves.isEmpty() ? Move.of(MoveType.PASS) : scriptedMoves.remo
    }
}
```

- CatanCoreEngine.java

```
package catan;

import java.util.ArrayList;
import java.util.List;
import java.util.Map;
import java.util.Set;

public final class CatanCoreEngine {
    private final List<CatanAgent> agents;
    private final Map<Integer, Integer> victoryPoints;
    private int turn;

    public CatanCoreEngine(List<CatanAgent> agents) {
        if (agents.size() != 4) {
            throw new IllegalArgumentException("Expected 4 agents");
        }
        this.agents = new ArrayList<>(agents);
        this.victoryPoints = Map.of(0, 0, 1, 0, 2, 0, 3, 0);
        this.turn = 0;
```

```
        for (int i = 0; i < this.agents.size(); i++) {
            this.agents.get(i).init(i);
        }
    }

    public Move runSingleTurn() {
        int currentPlayer = turn % agents.size();
        Set<MoveType> legalThisTurn = Set.of(MoveType.PASS, MoveType.BUILD_ROAD, Mov
        GameState state = new GameState(turn, currentPlayer, legalThisTurn, victoryP

        CatanAgent agent = agents.get(currentPlayer);
        Move chosen = agent.chooseMove(state);

        Move applied = state.isLegal(chosen) ? chosen : Move.of(MoveType.PASS, "ille
        turn++;
        return applied;
    }
}
```

- GameState.java

```
package catan;

import java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.Set;

public final class GameState {
    private final int turn;
    private final int currentPlayerId;
    private final Set<MoveType> legalMoveTypes;
    private final Map<Integer, Integer> victoryPoints;

    public GameState(int turn, int currentPlayerId, Set<MoveType> legalMoveTypes, Ma
        this.turn = turn;
        this.currentPlayerId = currentPlayerId;
        this.legalMoveTypes = Set.copyOf(legalMoveTypes);
        this.victoryPoints = Map.copyOf(victoryPoints);
    }

    public int turn() {
        return turn;
    }
```

```
    public int currentPlayerId() {
        return currentPlayerId;
    }

    public Set<MoveType> legalMoveTypes() {
        return Collections.unmodifiableSet(legalMoveTypes);
    }

    public Map<Integer, Integer> victoryPoints() {
        return Collections.unmodifiableMap(victoryPoints);
    }

    public boolean isLegal(Move move) {
        return legalMoveTypes.contains(move.type());
    }

    public List<Integer> possibleRobberTargets() {
        return victoryPoints.keySet().stream()
                .filter(id -> id != currentPlayerId)
                .toList();
    }
}
```

- Move.java

```
package catan;

import java.util.Objects;

public final class Move {
    private final MoveType type;
    private final String payload;

    private Move(MoveType type, String payload) {
        this.type = Objects.requireNonNull(type);
        this.payload = payload == null ? "" : payload;
    }

    public static Move of(MoveType type) {
        return new Move(type, "");
    }

    public static Move of(MoveType type, String payload) {
        return new Move(type, payload);
    }
```

```
    public MoveType type() {
        return type;
    }

    public String payload() {
        return payload;
    }

    @Override
    public String toString() {
        return type + (payload.isEmpty() ? "" : "(" + payload + ")");
    }
}
```

- MoveType.java

```
package catan;

public enum MoveType {
    PASS,
    BUILD_ROAD,
    BUILD_SETTLEMENT,
    BUILD_CITY,
    OFFER_TRADE,
    ACCEPT_TRADE,
    REFUSE_TRADE,
    COUNTER_TRADE,
    PLAY_DEVELOPMENT,
    MOVE_ROBBER
}
```

- Demo.java

```
package catan;

import java.util.List;

public final class Demo {
    private Demo() {
    }

    public static void main(String[] args) {
        CatanAgent a0 = new PolicyBackedCatanAgent(new ScriptedPolicy(List.of(
                Move.of(MoveType.BUILD_ROAD, "edge-3")
```

```
            )));
            CatanAgent a1 = new PolicyBackedCatanAgent(new ScriptedPolicy(List.of(
                    Move.of(MoveType.BUILD_CITY, "v-11")
            )));
            CatanAgent a2 = new PolicyBackedCatanAgent(new ScriptedPolicy(List.of(
                    Move.of(MoveType.OFFER_TRADE, "2 wool for 1 ore")
            )));
            CatanAgent a3 = new PolicyBackedCatanAgent(new ScriptedPolicy(List.of()));

            CatanCoreEngine engine = new CatanCoreEngine(List.of(a0, a1, a2, a3));

            for (int i = 0; i < 6; i++) {
                Move move = engine.runSingleTurn();
                System.out.println("Turn " + i + ": " + move);
            }
        }
    }
```

Key point: the engine never knows whether a move came from a stub, heuristic, or RL policy. That behavior is injected.

## 3.3 (c)

A common solution from a developer unfamiliar with these techniques is to place decision logic directly inside the engine, e.g., `if(agentType == RL) ... else ...`, or to hardcode random/heuristic behavior in core classes.

Problems with that approach and benefits of mine:

- **High coupling**: rule code and behavior code become tangled. In the current design, the core only sees `CatanAgent`.
- **Poor testability**: non-deterministic behavior makes core tests flaky. Scripted policies make tests reproducible.
- **Open/Closed violation**: every new agent type forces edits to core logic. With injection, new policies are added without touching engine code.
- **Higher maintenance risk**: bugs in learning code can destabilize rule validation. Separation keeps failures localized and easier to debug.

## 3.4 (d)

I would communicate to:

- technical peers (design session + code walkthrough),
- tech lead/manager (delivery and risk argument), and

- QA/testing members (how deterministic doubles improve test quality).

How and what I would communicate:

- short architecture note (problem, decision, tradeoffs, examples),
- live demo with the implemented files showing swappable policies,
- test strategy showing deterministic agent stubs for core-rule tests,
- team conventions: "core depends on interfaces, behavior injected".

Expectations:

- shared vocabulary (DI, test double, strategy),
- more stable and faster tests,
- easier onboarding and safer extension to RL agents.

With respect to the CEAB engineering practice: this reflects disciplined design, verification mindset, communication, professionalism, and continuous process improvement. It applies software engineering knowledge to reduce risk while improving quality and maintainability.

# 4 Task 4: Test-Driven Development(TDD)

## 4.1 (a)

The following small initial unit test suite for `divide()` can be used to verify basic functionality and sign semantics:

- `testDividePositiveNumbers`: verifies normal behaviour (`5.0 / 2.0 = 2.5`)

  ```
  private static void testDividePositiveNumbers(Calculator calculator) {
  assertDoubleEquals(2.5, calculator.divide(5.0, 2.0), 1e-9,
          "Expected 5.0 / 2.0 to be 2.5");
  }
  ```

- `testDivideNegativeByPositive`: verifies sign semantics (`-9.0 / 3.0 = -3.0`)

  ```
  private static void testDivideNegativeByPositive(Calculator calculator) {
          assertDoubleEquals(-3.0, calculator.divide(-9.0, 3.0), 1e-9,
                  "Expected -9.0 / 3.0 to be -3.0");
      }
  ```

The following helper methods should also be implemented to support the tests:

```
private static void assertDoubleEquals(double expected, double actual, double tolerance,
```

```
    if (Math.abs(expected - actual) > tolerance) {
        throw new AssertionError(message + ", got " + actual);
    }
}

private static void assertThrows(Class<? extends Throwable> expectedException,
                                 ThrowingRunnable action,
                                 String message) {
    try {
        action.run();
        throw new AssertionError(message + ", but no exception was thrown");
    } catch (Throwable t) {
        if (!expectedException.isInstance(t)) {
            throw new AssertionError(message + ", got " + t.getClass().getSimpleName());
        }
    }
}

@FunctionalInterface
private interface ThrowingRunnable {
    void run() throws Throwable;
}
```

These tests define the first iteration's target behaviour before implementing full edge-case handling.

## 4.2 (b)

The basic functionality can be implemented as follows to make the aforementioned tests pass:

```
package calculator;

public final class Calculator {
    public double divide(double numerator, double denominator) {
        return numerator / denominator;
    }
}
```

- method `divide(double numerator, double denominator)`
- returns `numerator / denominator` for valid normal inputs

The tests were then executed from the command line and were confirmed to pass.

## 4.3 (c)

After iteration 1, the functionality is not complete. The first tests only validate nominal arithmetic behaviour and one sign case. Missing behaviour becomes obvious when compared with code one would write non-TDD in one pass, where developers often immediately think about exceptional cases.

Important missing behaviours:

- division by zero (`0.0` and `-0.0`)
- invalid floating input such as `NaN`
- explicit and documented error semantics (which exception and why)

TDD made this visible quickly: once minimal tests pass, the next step is to identify what behaviour is still unspecified. If a behaviour is not in tests, it is effectively not part of the contract yet.

## 4.4 (d)

The test suite was then expanded by adding a second TDD iteration focused on edge cases:

- `testDivideByZeroThrows`: expects `ArithmeticException` for denominator `0.0`

  ```
  private static void testDivideByZeroThrows(Calculator calculator) {
          assertThrows(ArithmeticException.class,
                  () -> calculator.divide(10.0, 0.0),
                  "Expected division by +0.0 to throw ArithmeticException");
      }
  ```

- `testDivideNegativeZeroThrows`: expects `ArithmeticException` for denominator `-0.0`

  ```
  private static void testDivideNegativeZeroThrows(Calculator calculator) {
      assertThrows(ArithmeticException.class,
              () -> calculator.divide(10.0, -0.0),
              "Expected division by -0.0 to throw ArithmeticException");
  }
  ```

- `testDivideNaNThrows`: expects `IllegalArgumentException` if an argument is `NaN`

  ```
  private static void testDivideNaNThrows(Calculator calculator) {
          assertThrows(IllegalArgumentException.class,
                  () -> calculator.divide(Double.NaN, 2.0),
                  "Expected NaN input to throw IllegalArgumentException");
      }
  ```

These tests extend the contract from “works on normal numbers” to “fails safely on invalid input.”

## 4.5 (e)

The `divide()` method can then be updated to satisfy the expanded suite in the following way:

```
package calculator;

public final class Calculator {
    public double divide(double numerator, double denominator) {
        if (Double.isNaN(numerator) || Double.isNaN(denominator)) {
            throw new IllegalArgumentException("NaN is not supported");
        }

        if (denominator == 0.0) {
            throw new ArithmeticException("Cannot divide by zero");
        }

        return numerator / denominator;
    }
}
```

- validate inputs and reject `NaN` with `IllegalArgumentException`
- reject zero denominator (including `-0.0`) with `ArithmeticException`
- keep normal arithmetic behaviour unchanged

## 4.6 (f)

I would use TDD in contexts where correctness and regression safety are important: utility/-domain logic, business rules, parsers, and modules with clear input/output contracts.

I would use it less strictly for heavy UI prototyping or exploratory tasks where requirements change rapidly by design.

Element I liked most:

- it forces explicit behaviour contracts and prevents overbuilding

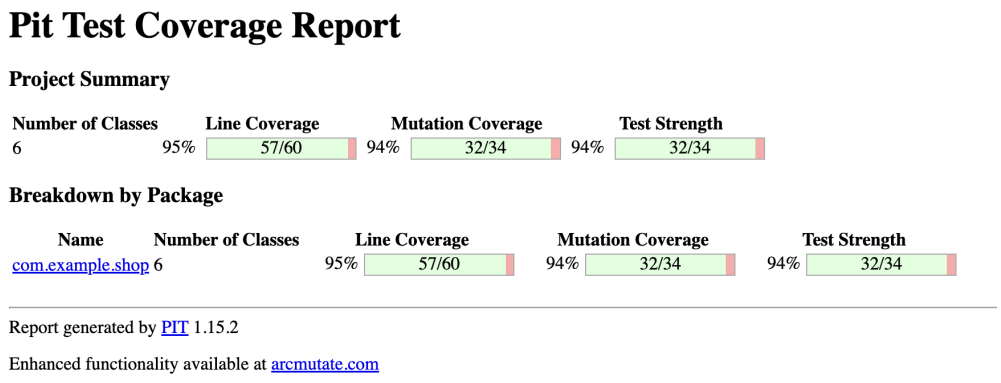
Hardest element to manage:

- selecting the right test granularity and not overfitting implementation details while still covering edge cases

Overall, this exercise showed the TDD cycle clearly across two iterations: start with minimal behaviour, then expand the contract with edge cases and update implementation accordingly.

# 5 Task 5: Test coverage and AI

## 5.1 a)



**Pit Test Coverage Report**

**Project Summary**

| Number of Classes | Line Coverage | | Mutation Coverage | | Test Strength | |
|---|---|---|---|---|---|---|
| 6 | 95% | 57/60 | 94% | 32/34 | 94% | 32/34 |

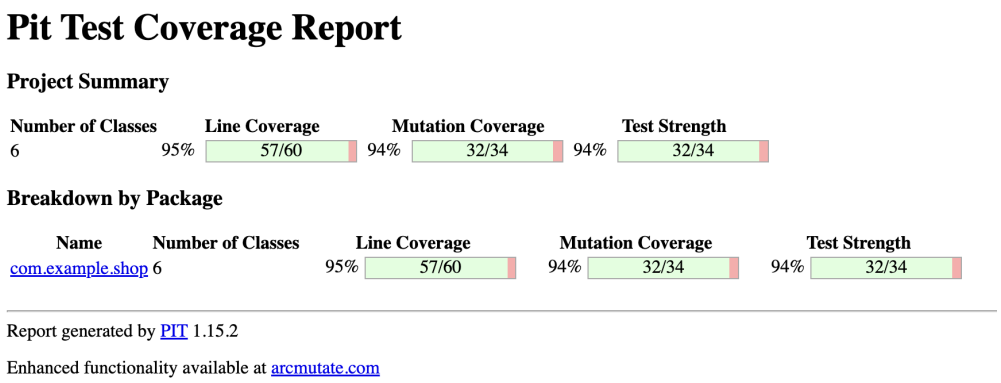
**Breakdown by Package**

| Name | Number of Classes | Line Coverage | | Mutation Coverage | | Test Strength | |
|---|---|---|---|---|---|---|---|
| com.example.shop | 6 | 95% | 57/60 | 94% | 32/34 | 94% | 32/34 |

Report generated by PIT 1.15.2

Enhanced functionality available at arcmutate.com

Figure 1: Branch Coverage Report



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Report generated by PIT 1.15.2

Enhanced functionality available at arcmutate.com

Figure 2: Mutation Testing Report

I queried Gen AI with this prompt to generate a reasonable amount of test cases for the e-commerce order processing system:

**"Generate a reasonable amount of JUnit test cases for all the classes within the given e-commerce order processing system"**

With this prompt, I was able to generate an initial Jnit suite for the order-procesing classes. To measure coverage, I used Jacoco branch coverage and PIT mutation testing to get a mutation score. The generated tests achieved 88.89% branch coverage and a mutation score of 94%. The branch coverage indicates that most of the decision outcomes in the code were exercised by the tests, while the high mutation score indicates that the tests were effective at killing mutants, suggesting the test suite is good at catching faults.

## 5.2 b)

**task5-shop-tests**

| Element | Missed Instructions | Cov. | Missed Branches | Cov. | Missed | Cxty | Missed | Lines | Missed | Methods | Missed | Classes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| com.example.shop | | 100% | | 100% | 0 | 33 | 0 | 64 | 0 | 18 | 0 | 7 |
| Total | 0 of 248 | 100% | 0 of 27 | 100% | 0 | 33 | 0 | 64 | 0 | 18 | 0 | 7 |

Figure 3: Branch Coverage Report

**Pit Test Coverage Report**

**Project Summary**

| Number of Classes | Line Coverage | | Mutation Coverage | | Test Strength | |
|---|---|---|---|---|---|---|
| 6 | 100% | 60/60 | 97% | 33/34 | 97% | 33/34 |

**Breakdown by Package**

| Name | Number of Classes | Line Coverage | | Mutation Coverage | | Test Strength | |
|---|---|---|---|---|---|---|---|
| com.example.shop | 6 | 100% | 60/60 | 97% | 33/34 | 97% | 33/34 |

Report generated by PIT 1.15.2

Enhanced functionality available at arcmutate.com

Figure 4: Mutation Testing Report

To drive the branch coverage past 90% and the mutation score closer to 100%, I worked in conjunction with generative AI to develop addition test cases.

Since the AI was not good at finding what specific paths were not covered, the task fell on me (the human) to investigate the coverage report and find the missing paths that the tests were not covering. Once I found a missing path, I would prompt the AI to generate the test case(s) that would cover that path. I would then run the coverage report again and repeat the process until I was satisfied with the coverage and mutation score.

It is clear that I (the human) played the role of investigating the coverage report and analyzing the code to find gaps in the test suite, while the AI played the role of generating the test cases once I found the problem areas. This back and forth process allowed me to both apply my understanding of the code and testing principles while leveraging the AI's ability to rapidly generate test cases with my guidance.

It took me about 45 minutes to reach 100$ branch coverage, and a mutation score of 97%.

## 5.3 c)

My key takeaways from this exercise are that generative AI is a powerful tool for rapidly generating large amounts of code, including test cases, but fall short in identifying specific

coverage gaps.

The tedious part of coding: writing the boilerplate, ensuring correct syntax, remembering certain methods or libraries, is done extremely quickly by AI and is usually correct. *However*, the "engineering" part of software development: understanding the system, analyzing coverage reports, finding gaps in the test suite, and subsequently creating effective test cases to fill those gaps, still depends heavily on a human's understanding. This is where the strengths of AI and humans both differ and complement eachother. AI can handle generating large amounts of code quickly, but human cognition is needed to guide the process, analyze the coverage reports, and to show the AI where to focus its efforts.

This exercise did a great job of showcasing that a human eye and human critical thinking is still needed when designing a test suite and reaching coverage goals, even when using state-of-the-art generative AI.

## 5.4 d)

If I had $1M to invest in the next generation of AI-enabled, human-in-the-loop testing tools, one of the top features of that tool would be an Always-on conversational testing copilot. It would work as a voice-enabled (or text) assistant that continuously monitors testing activities, coverage reports, and code changes. It would give instant feedback from code changes like (the change you just made in *PaymentValidator* dropped your branch coverage from 100% to 85%, and could then take voice input on what to do next ("show me the new coverage report", "explain why the coverage dropped", "generate test cases that covers so and so path, etc). This would allow for a much more seamless experience when designing tests where the human acts as the "conductor" of the testing process, while the AI acts as the "copilot" which follows the human's lead and provides instant feedback. The AI would have to be able to process the voice input quickly and accurately which would likely take a big chunk of the investment, but the end result would be an extremely powerful tool that works akin to a super fast, always-on, pair programmer.

The biggest challenges of widespread adoption would almost certaintly be around trust. Many teams would not be willing to give an AI a high level of access to their codebase, much less trust it to execute their instructions around code changes and test generation correctly. Furthermore, since generative AI is still a black box, teams that work with software that controls critical systems (medical, aerospace, etc) would likely face regulatory hurdles around giving an AI access to a codebase that has real world consequences.

Overcoming these challenges would likely require education about how the AI works, as well as strict access controls to ensure the AI cannot overstep its bounds and any changes it makes both requires human approval and are fully reversible.